ეკონომიკური ინჟინერინგი ფარმაციაში

WET COUGH AND SOME EFFECTIVE EXPECTORANT MEDICINAL PLANTS

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ABSTRACT

A cough, also known as tussis, is a voluntary or involuntary act that clears the throat and breathing passage of foreign particles, microbes, irritants, fluids, and mucus; it is a rapid expulsion of air from the lungs. There are many types of drugs that are used to suppress cough and are often prescribed in combination. Traditionally cough is classified as either productive, i.e. producing mucus usually with expectoration, or nonproductive (dry). Expectorants are the herbs that facilitate or accelerate the removal of bronchial secretions from the bronchi and trachea. There are two general categories of expectorant herbs: stimulating and relaxing. Most stimulating expectorants contain alcaloids, saponins, or volatile oils.

Thyme (Thymus Vulgaris L) assissts productive cough and is used for treatment of bronchitis, whooping cough, asthma and catarrh or inflammation of the upper respiratory tract. Ivy (Hedera helix) is used in the treatment of catarrhal disorders of the upper respiratory tract and in the symptomatic treatment of chronic inflammatory bronchial diseases. The HMPC (the European Medicines Agency's (EMA) committee) concluded that Ivy leaf preparations can be used as an expectorant for chesty coughs.Blue gum (Eucaliptus globulus) oil, Eucaliptol, irritates the mucous membrane causing an increase of bronchial secretion (for defense) and facilitate the expulsion of phlegm and mucus through the cough. According to The HMPC, Anise (Pimpinela anisum) oil, obtained by steam distillation of the dry ripe fruits, can be used as an expectorant for coughs associated with colds.

Conclusion. The following article summerizes folk medicine and scientific data on some expectorant plants that can successfully be used for productive cough treatment. Some well tested receipts from folk medicines have to be actively consumed together with synthesized medications

Key words: Cough, expectorant, Thyme (Thymus Vulgaris L), Ivy (Hedera helix),Blue gum (Eucalyptus globulus), Anise (Pimpinela anisum).

სველი ხველა და ზოგიერთი ეფექტური ამოსახველებელი სამკურნალო მცენარე

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რეზიუმე

ხველა წარმოადგენს ნებსით თუ უნებლიე, რეფლექსურ, ქმედებას, რითაც ორგანიზმი ყელსა და სასუნთქ სისტემას ასუფთავებს მასში მოხვედრილი უცხო სხეულების, მიკრობების, გამღიზიანებლების, სითხეებისა და ლორწოსგან. ხველა არის ფილტვებიდან ჰაერის სწრაფი გამოდევნა. არსებობს ხველის საწინააღმდეგო ბევრი მედიკამენტი, რომლებიც ხშირად კომბინაციაში გამოიყენება.

ხველა იყოფა პროდუქტიულ, ანუ ნახველის გამოყოფით მიმდინარე, ან არაპროდუქტიულ (მშრალ) ხველად. ამოსახველებელი მცენარეები ხელს უწყობენ ან აჩქარებენ ბრონქული სეკრეტის გამოდევნას ბრონქებიდან და ტრაქეიდან. არსებობს ორი ტიპის ამოსახველებელი მცენარეული საშუალება: მასტიმულირებელი და დამამშვიდებელი. მასტიმულირებელი ამოსახველებლების უმეტესობა შეიცავს ალკალოიდებს, საპონინებსა და ეთერზეთებს.

სტატიაში თავმოყრილია მონაცემები ფიტოთერაპიაში გამოყენებულ ზოგიერთ ეფექტურ ამოსახველებელ მცენარეულ საშუალებაზე.

ბეგქონდარა (Thymus Vulgaris L) ეხმარება პროდუქტიულ ხველას და გამოიყენება ბრონქიტის, ყივანახველის, ასთმის, კატარისა და ზოგადად, ზედა სასუნთქი გზების ანთებითი დაავადებების დროს. სურო (Hedera helix) ეფექტურია ზედა სასუნთქი გზების კატარისა და ბრონქების მწვავე ანთებითი დაავადებების სიმპტომური მკურნალობისას. მცენარეული სამკურნალო პროდუქტების კომიტეტის

(წამლის ევროპული სააგენტოს კომიტეტი) დასკვნით, სუროს ფოთლების პრეპარატები ეფექტური ამოსახველებელი საშუალებაა მკერდისმიერი ხველისას. ევკალიპტის (Eucaliptus globulus) ზეთი, ევკალიპტოლი, აღიზიანებს რა ლორწოვან ზედაპირს, ზრდის ბრონქული სეკრეტის გამოყოფას (დაცვის მიზნით) და ხელს უწყობს ხველებით ლორწოს გამოდევნას. მცენარეული სამკურნალო პროდუქტების კომიტეტის დასკვნის მიხედვით, ანისულის (Pimpinela anisum) ზეთი, რომელიც მიიღება მისი მწიფე, მშრალი ნაყოფის წყლის ორთქლით გამოხდისას, გამოიყენება როგორც ამოსახველებელი საშუალება გაცივებით გამოწვეული ხველებისას.

დასკვნა. სტატიაში თავმოყრილია ხალხურ მედიცინასა და სამეცნიერო კვლევებში არსებული მონაცემები ზოგიერთ ამოსახველებელ მცენარეულ საშუალებაზე, რომლებიც წარმატებით გამოიყენება პროდუქტიული ხველისას. ხალხურ მედიცინაში კარგად გამოცდილი ზოგიერთი რეცეპტი აქტიურად უნდა იქნას გამოყენებული თანამედროვე სამკურნალო საშუალებებთან ერთად.

საკვანძო სიტყვები: ხველა, ამოსახველებელი, ბეგქონდარა (Thymus Vulgaris L), სურო(Hedera helix), ევკალიპტი (Eucalyptus globulus), ანისული (Pimpinela anisum).

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A cough, also known as tussis, is a voluntary or involuntary act that clears the throat and breathing passage of foreign particles, microbes, irritants, fluids, and mucus; it is a rapid expulsion of air from the lungs. During common cold pharyngeal mucosa dries out and the cough receptors in the throat become more irritable.

This mucous secretions help to protect the respiratory tract from all kinds of irritants by trapping and flushing out smoke particles, bacteria, and viruses. Any cough that lasts more than a few days, does not respond to treatment, or produces blood should be investigated further, as may be a sign of serious organic disease.

Acute inflammatory conditions of the respiratory system are primarily treated with mucilage-rich demulcents, which soothe inflamed tissue. Stimulants, saponin-containing expectorants are best used for subacute or chronic bronchitis, for which active expectoration is indicated. However, any irritation in chronic bronchitis will indicate the need for increased demulcent action (7).

There are many types of drugs that are used to suppress cough and are often prescribed in combination. Before dealing with the particular type of drug used, it is important to consider briefly the nature of cough production, its role in disease and desirability of suppressing it (1) In most countries, these pharmaceutical preparations are sold as over-the-counter (OTC) drugs, a category that includes the following therapeutic classes: cough suppressants and expectorants (2). Cough is thought to be caused by a reflex. It occurs due to stimulation of mechano-or chemoreceptor in throat, respiratory passage or stretch receptor in the lungs(3). The sensitive receptors are located in the bronchial tree, particularly in the junction of the trachea. These receptors can be stimulated mechanically or chemically e.g. by inhalation of various irritants than nerve impulses activate the cough center in the brain (4).

Traditionally cough is classified as either productive, i.e. producing mucus usually with expectoration, or nonproductive (dry) (5). They help to raise secretions from the respiratory passages (6).

Many respiratory conditions are characterised by abnormal mucus (catarrh) that can narrow airways. This abnormal mucus may be thick and tenacious and hence very diffcult to clear from the airways. By clearing abnormal mucus or by changing its character and making it more demulcent, expectorants can allay cough and are therefore antitussive(8).

Expectorants are the herbs that facilitate or accelerate the removal of bronchial secretions from the bronchi and trachea. There are two general categories of expectorant herbs: stimulating and relaxing (7).

Stimulating expectorants act in different ways. For any given stimulating expectorant, either or both of the following mechanisms may work:

1. Irritation of bronchioles stimulates the expulsion of any material present. This irritation initiates a reflex response that causes the body to cough.
2. Liquefaction of viscid sputum encourages clearing by coughing. Sputum is moved upword from the lungs by the fine hairs of the ciliated epithelium that lines the bronchole tubes, and reduced sputum viscosity facilitates the transport.

Most stimulating expectorants contain alcaloids, saponins, or volatile oils. However, not all chemicals in these groups or plants containing these constituents have expectorant properties. The particular form of stimulation offered by this group of expectorants makes them relevant for productive coughs, in which sputum should be removed from the airways (7).

Relaxing expectorants may also act by reflex, but here the herbs work to soothe bronchial spasm and loosen mucus secrections. This loosening occurs because relaxing expectorants facilitate the production of a less viscous mucous secretion, which helps lift up stiker material from below. This makes relaxing expectorants useful for dry, irritating coughs. You will notice that this action is similar in some respects to that of demulcents, and both actions owe a lot to their content of mucilage and occasionally volatile oils (7).

Indications for expectorants are:

- Cough linked to bronchial congestion, especially where mucus is thick and tenacious or where there is

- unproductive cough;
- Bronchitis, emphysema;
- Productive cough associated with cold;
- Profuse catarrhal conditions;
- Dry cough.

Expectorants are best taken in hot infusions or as tinctures or fluid extracts, before food.

Contraindications for expectorants include:

- dry and irritable conditions of the lungs;
- asthma;
- young children;
- dyspeptic conditions;
- gastro-oesophageal reflux.

This article discusses some most effective expectorant plants Thyme (Thymus Vulgaris L), Ivy (Hedera helix), Blue gum (Eucalyptus globulus), and Anise (Pimpinella anisum).

Thyme (Thymus Vulgaris L) extract in combination with other herbs is available in a number of commercial products to treat cough and/or acute bronchitis. It has expectorant, spasmolitic, antibacterial, antifungal, antioxidant, and anti-inflammatory properties. According to clinical trials, Thyme assissts productive cough. Traditionnal folk medicines the herb has been using for bronchitis, whooping cough, asthma and catarrh or inflammation of the upper respiratory tract (8).

Thymus vulgaris is a member of the Lamiaceae (Labiatae or mint) family, native to the western Mediterranean and southern Italy, but widely cultivated. It is an evergreen sub-shrub growing up to 25cm, with a branched stem. The sessile leaves vary from elliptic to linear or diamond-shaped. The small pink to lilac flowers appear in whorls in the upper leaf axils and have a tube-like calyx and tubular corolla with a three lobed lower lip. The fruit consists of a smooth, dark-coloured nutlet and the roots are robust. The whole plant has an aromatic fragrance (15).

Figure 1. Thymus vulgaris

Figure 2. Thymus vulgaris leaves

Figure 3. Thymus vulgaris flowers

Figure 4. Thymus vulgaris fruits

For medical purposes there are used the herb and its oil. The herb has very pleasant aromatic odour and camphor like taste. The oil is tansparent with yellow or brownish color and pleasant aromatic odour (29).

Key constituents include essential oil (1% to 2.5%) containing antimicrobial and antioxidant (12)(13), phenols, predominantly thymol and/or carvacrol (20) (up to 51% and 4%, respectively, of the essential oil, but varying considerably with the many chemotypes of thyme), and their corresponding monoterpene hydrocarbon precursors (p-cymene and gamma-terpinene),-together with thymol methyl ether (1.5% to 2.5%).

In a double blind, randomised study, 60 patients with acute respiratory infection and productive cough received either syrup of thyme or bromhexine for a period of 5 days. There was no signifcant difference between the two groups based on self-reported symptom relief and both groups made similar gains (8).

Typical adult dosage ranges are:

- 3 to12g/day of dried aerial parts or by infusion
- 2 to 6mL/day of a 1:2 liquid extract or equivalent in tablet or capsule form
- 6 to 18mL/day of a 1:5 tincture.
- Topical use: 5% infusion as a gargle or mouthwash

In folk medicien, 15 g crude drug is boiled in 1 cup of water and taken 1 table spoon 3 times a day and for decoction to be prepared, crude drug is boiled in 200 ml water in ratio 1:10, then cooled at room temperature and taken 1-2 table spoon 3.-5 times a day (29).

Thyme oil should not be applied near the mouth and nose of infants and young children as it can provoke a dangerous reflex breathing spasm (21). Overdose provokes vomiting. It is not recommended during pregnancy and for patients with renal and hepatic diseases (29). Otherwise no adverse effects from thyme are expected.

Ivy (Hedera helix) is the only saponin-containing herb that is widely prescribed as a single –herb preparation. It is used in the treatment of catarrhal disorders of the upper respiratory tract and in the symptomatic treatment of chronic inflammatory bronchial diseases. The HMPC (Committee on Herbal Medicinal Products) concluded that Ivy leaf preparations can be used as an expectorant (a medicine that helps to bring up phlegm) for productive (chesty) coughs. Ivy leaf medicines should only be used in adults, adolescents and children from the age of 2 years (30).

Hedera helix, the common ivy, English ivy, European ivy, or just ivy, is a species of flowering plant of the ivy genus in the family Araliaceae, native to most of Europe and western Asia. A rampant, clinging evergreen vine, it is a familiar sight in gardens, waste spaces, and wild areas, where it grows on walls, fences, tree trunks, etc.

Hedera helix is an evergreen climbing plant, growing to 20–30 m high where suitable surfaces (trees, cliffs, walls) are available, and also growing as groundcover where no vertical surfaces occur. It climbs by means of aerial rootlets with matted pads which cling strongly to the substrate (Figure 5).

The leaves are alternate, 50–100 mm long, with a 15–20 mm petiole; they are of two types, with palmately five-lobed juvenile leaves on creeping and climbing stems, and unlobed cordate adult leaves on fertile flowering stems exposed to full sun, usually high in the crowns of trees or the top of rock faces.

The flowers are produced from late summer until late autumn, individually small, in 3-to-5 cm-diameter umbels, greenish-yellow, and very rich in nectar, an important late autumn food source for bees and other insects. The fruit are purple-black berries 6–8 mm in diameter, ripening in late winter.

Figure 5. Hedera helix

Figure 6. Hedera helix flowers

Ivy berries are somewhat poisonous to humans, but ivy extracts are part of current cough medicines. In the past, the leaves and berries were taken orally as an expectorant to treat cough and bronchitis (25). In 1597, the British herbalist John Gerard recommended water infused with ivy leaves as a wash for sore or watering eyes (26). The leaves can cause severe contact dermati-

Figure 7. Hedere Helix fruits

tis in some people.

Previous studies showed that the Hedera helix extract contains alpha- and beta-hederin (α-hederin and β-hederin), falcarinol, didehydrofalcarinol, rutin, caffeic acid, chlorogenic acid, emetine, nicotiflorin, hederasaponin B and hederacoside C. However, only three extracted components were detectable more than 1.5% in the European ivy leaves (hederacoside C 15.69%, chloro-genic acid 2.07%, and rutin 1.62%). Other components were detectable in very few amounts (< 1%) or not detectable in some studies (27).

The average recommended daily dose of Hedera helix is 0.3 g of the dried herb or an equivalent dose of extract.

Four controlled studies in patients with chronic obstructive bronchitis have demonstrated the therapeutic efficacy of a water-ethanol extract (30 % ethanol, herb-to-extract ratio 5-7:1, brand name Prospan) in adults (24) as well as children. One of these studies was a prospective double blind study in 99 women and men 25-70 years of age diagnosed with chronic obstructive bronchitis. Four weel's treatment with a daily dose of approximately 60 mg of ivy extract (equivalent to about 400 mg of crude drug) was compared with 90 mg of ambroxol.

Figure 8. Prospan, Hedera helix leaves extract

Side effects affecting the stomach and gut such as nausea, vomiting and diarrhoea, and allergic reactions such as hives, skin rash and difficulty breathing have been reported with ivy leaf medicines, although their frequency is not known. Ivy leaf medicines must not be taken by people who are hypersensitive (allergic) to ivy leaf or to other plants of the ivy family (Araliaceae). Ivy leaf medicines must not be given to children under 2 years of age because of the risk of worsening respiratory symptoms when using cough medicines at this age (36)

Eucalyptus globulus, commonly known as southern blue gum(28) or blue gum, is a species of flowering plant in the family Myrtaceae. It is a tall, evergreen tree endemic to southeastern Australia. There are four subspecies, each with a different distribution across Australia, occurring in New South Wales, Victoria and Tasmania. The subspecies are the Victorian blue gum, Tasmanian blue gum, Maiden's gum, and Victorian eurabbie.

Eucalyptus globulus is a tree that typically grows to a height of 45 m but may sometimes only be a stunted shrub, or alternatively under ideal conditions can grow as tall as 90–100 m. The bark is usually smooth, white to cream-coloured but there are sometimes slabs of persistent, unshed bark at the base.

Figure 9. Eucaliptus globulus

Figure 10. Eucaliptus globulus bark

Juvenile leaves are mostly arranged in opposite pairs up to 150 mm long and 105 mm wide. The flower buds are arranged singly or in groups of three or seven in leaf axils. This Eucalyptus has woody ftuits.

Figure 11. Eucaliptus leaves

Figure 12. Eucaliptus flowers

Figure 13. Eucaliptus fruits

Oil from the eucalyptus leaves, known as eucaliptol, appears in many over-the-counter cough and cold products to relieve congestion.Approximately 7 % of eucalyptus oil is comprised of cineol. Eucaliptol is colorless or yellowish liquid with a specific, pleasant smell of cineole and a sharp, cooling taste and possesses antispasmodic, sectretagogic (promotes secretion), sectretolytic (breaks up secretions (especially phlegm)), antimicrobial, and fungicidal (toxic to or destroys fungi) properties. Eucaliptol causes irritation in the mucous membrane, causing an increase of bronchial secretion (for defense) and facilitate the expulsion of phlegm and mucus through the cough. Recommended dose of eucalyptus oil is an average 0.3-0.5 g daily for upper respiratory infections.

Eucaliptol is widely used in pharmaceutical products like Pectussin tablets (Tabulettae Pectusinum), Inhalipt oral spray(Inhalyptum), Eucatol capsules (Guttae Eucatolum), Kofanol tablets, Tetesept eucaliptol capsules, etc.

Figure 14. Inhalipt oral spray

Figure 15. Eucatol capsules

In Georgian folk medicine decoction of Eurcalitus leaves has been used as an antiseptic and axpectrant remedy and was prepared as follows: 15 g of crude drug is added 200 ml water and boiled for half an hour. Decoction should be refilled to 200 ml again with boiling water. The product should be taken 50 ml /three times a day before meal (29) (figure 16-17).

Adverse effects of eucaliptol include rare instance of stomach upset following internal use, and external use occasionally causes hypersensitivity reactions on the skin. There is almost no reports on cineol toxicity (8).

Anise (Pimpinella anisum), also called aniseed or rarely anix is a flowering plant in the family Apiaceae native to the eastern Mediterranean region and Southwest Asia. (31).

Figure 16. Anise, Pimpinella anisum

Figure 17. Anise flowers

The name “anise” is derived via old French from the Latin words anisum or anethum from Greek referring to dill. (32) An obsolete English word for anise is anet.

Anise is an herbaceous annual plant growing to 60–90 centimetres or more. The leaves at the base of the plant are simple, 1–5 cm long and shallowly lobed, while leaves higher on the stems are feathery or lacy, pinnate, divided into numerous small leaflets (33).

Both leaves and flowers are produced in large, loose clusters. The flowers are either white or yellow, approximately 3 millimetres in diameter.

The fruit is a dry oblong and curved schizocarp, 4–6 mm long, usually called “aniseed” (33).

Figure 18. Aniseeds

Figure 19. Anise oil

Anise was first cultivated in Egypt and the Middle East, and was brought to Europe for its medicinal value. It has been cultivated in Egypt for approximately 4,000 years (34).

Anise oil is essential oil obtained from the ripe fruits of Pimpinela anisum. The HMPC conclusions only cover anise oil preparations which are obtained by steam distillation of the dry ripe fruits. Herbal medicines containing these anise oil preparations are usually available in solid or liquid forms to be taken by mouth. The HMPC concluded that the oil can be used as an expectorant (a medicine that helps bring up phlegm) for coughs associated with colds. Anise oil medicines should only be used in adults and should not be taken for longer than two weeks. (35)

The principal component (80-90%) of anise oil is anethole which is obtained from anise oil by freezing. Anise oil is clear , colorless liquid with a spicy odor and a sweet, aromatic taste that solidifies to a white crystalline mass when refrigerated. Anethole forms white crystals that melt at 20^0 C -22 ^{0} C. The expectorant effects of anise oil and anethole are presumably based on their ability to stimulate the ciliary activity of the bronchial epithelium. Anethole is rapidly absorbed from the gastrointestinal tract and is rapidly eliminated with the urine. Average daily oral dose is 3 g of crude drug and 0.3 g of essential oils.

Anise may have estrogen-like effects, so there’s some concern that the use of anise supplements may be potentially harmful to people with hormone-sensitive conditions, such as hormone-dependent cancers (breast cancer, uterine cancer, ovarian cancer), endometriosis, and uterine fibroids. Anise may also interact with certain medications including birth control pills, estrogen, and tamoxifen (37). Occasional allergic skin reactions have also been reported.

Having analysed folk medicine and scientific data on expectorants, we concluded that number of medicinal plants can successfully be used for productive cough treatment and well tested receipts from folk medicines have to be restored and consumed together with synthesized medications.

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